

```

40 C22 = add(subtract(add(M1, M3), M2), M0)
41
42 C = []
43
44 for i in range(mid):
45     C.append(C11[i] + C12[i])
46
47 for i in range(mid):
48     C.append(C21[i] + C22[i])
49
50 return C
51
52
53 A = []
54 B = []
55
56 n = int(input())
57
58 for i in range(n):
59     A.append(list(map(int, input().split())))
60
61 for i in range(n):
62     B.append(list(map(int, input().split())))
63
64 result = strassen(A, B)
65
66 for row in result:
67     print(*row)

```



```

4 5
6 5
7 8
58 57
59 60

```

...Program finished with exit code 0

```

1 def next_power_of_two(n):
2     p = 1
3
4     while p < n:
5         p = p * 2
6
7     return p
8
9
10 def strassen_count(n):
11     size = next_power_of_two(n)
12     count = 1
13
14     while size > 1:
15         count = count * 7
16         size = size // 2
17
18     return count
19
20
21 def standard_count(n):
22     return n * n * n
23
24
25 n = int(input())
26
27 print("Strassen Multiplications =", strassen_count(n))
28 print("Standard Multiplications =", standard_count(n))

```

4

```

Strassen Multiplications = 49
Standard Multiplications = 64

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```

```

60 AP = [[0] * size for _ in range(size)]
61 BP = [[0] * size for _ in range(size)]
62
63 for i in range(n):
64     for j in range(n):
65         AP[i][j] = A[i][j]
66         BP[i][j] = B[i][j]
67
68 CP = strassen(AP, BP)
69
70 return [row[:n] for row in CP[:n]]
71
72
73 n = int(input())
74
75 A = []
76 B = []
77
78 for i in range(n):
79     A.append(list(map(int, input().split())))
80
81 for i in range(n):
82     B.append(list(map(int, input().split())))
83
84 result = multiply(A, B)
85
86 for row in result:
87     print(*row)

```

input

```

7 8 9
30 36 42
66 81 96
102 126 150

```

...Program finished with exit code 0
Press ENTER to exit console.

```

57 C22 = add(subtract(add(M1, M3), M2), M0)
58
59 C = []
60
61 for i in range(mid):
62     C.append(C11[i] + C12[i])
63
64 for i in range(mid):
65     C.append(C21[i] + C22[i])
66
67 return C
68
69
70 n = int(input())
71
72 A = [[random.randint(1, 10) for j in range(n)] for i in range(n)]
73 B = [[random.randint(1, 10) for j in range(n)] for i in range(n)]
74
75 start = time.time()
76 standard_multiply(A, B)
77 standard_time = time.time() - start
78
79 start = time.time()
80 strassen(A, B)
81 strassen_time = time.time() - start
82
83 print("Standard Time =", standard_time)
84 print("Strassen Time =", strassen_time)

```

input

```

4
Standard Time = 3.1948089599609375e-05
Strassen Time = 0.00025010108947753906

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```

```

54
55     C = []
56
57     for i in range(mid):
58         C.append(C11[i] + C12[i])
59
60     for i in range(mid):
61         C.append(C21[i] + C22[i])
62
63     return C
64
65
66 n = int(input())
67 threshold = int(input())
68
69 A = []
70 B = []
71
72 for i in range(n):
73     A.append(list(map(int, input().split())))
74
75 for i in range(n):
76     B.append(list(map(int, input().split())))
77
78 result = strassen_hybrid(A, B, threshold)
79
80 for row in result:
81     print(*row)

```

input

```

1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```